

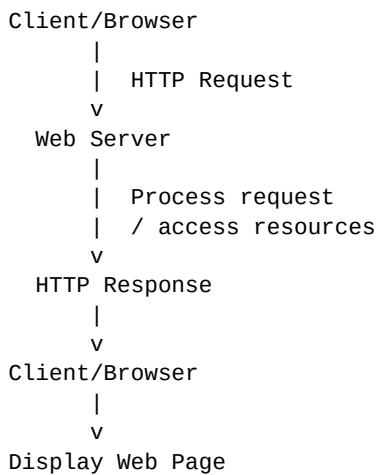
Frontend – Assignment 1 (Solved)

Based on the questions provided in the uploaded assignment.

Q1. HTTP Request–Response Cycle

The HTTP Request–Response Cycle is the process through which a client, usually a web browser, communicates with a web server to obtain a resource such as an HTML page, CSS file, image, or API data.

Major steps: (1) The user enters a URL or clicks a link. (2) The browser finds the server address using DNS. (3) The browser establishes a network connection with the server. (4) It sends an HTTP request containing the method, URL/path, headers, and sometimes a body. (5) The web server processes the request and may access databases or other services. (6) The server sends an HTTP response containing a status code, headers, and usually a response body. (7) The browser receives the response, interprets the HTML/CSS/JavaScript, and displays the web page.



Q2. Introduction to Web Technologies

1. HTML: HyperText Markup Language structures the content of a web page. It provides elements such as headings, paragraphs, links, images, forms, and tables.

2. CSS: Cascading Style Sheets controls presentation and layout, including colors, fonts, spacing, borders, positioning, and responsive design.

3. JavaScript: A programming language used to add behavior and interactivity to web pages, such as event handling, validation, dynamic content, and API communication.

4. Bootstrap: A frontend CSS/JavaScript framework that provides reusable components and a responsive grid system, helping developers build interfaces faster.

5. jQuery: A JavaScript library that simplifies DOM manipulation, event handling, animations, and some browser-related tasks.

6. React.js: A JavaScript library for building component-based user interfaces. It helps developers create reusable UI components and efficiently update the interface when data changes.

Q3. Client-Side vs. Server-Side Technologies

Client-side technologies run mainly in the user's browser. They control what the user sees and how the page behaves. Examples include HTML, CSS, JavaScript, and React.js.

Server-side technologies run on the web server. They handle business logic, authentication, database operations, and generation or delivery of data. Examples include Node.js, Java (Spring), Python (Django/Flask), and PHP.

Key difference: Client-side code executes on the client/browser, while server-side code executes on the server. Client-side work focuses on presentation and interaction; server-side work focuses on processing, data, security, and application logic.

Aspect	Client-Side	Server-Side
Runs on	Browser	Web server
Main purpose	UI and interaction	Logic and data processing
Examples	HTML, CSS, JavaScript, React	Node.js, Java, Python, PHP

Q4. Block-Level Elements in HTML

Block-level elements normally start on a new line and occupy the available horizontal space of their containing element. They are commonly used to structure sections of a page.

Characteristics include starting on a new line, generally taking the full available width, and allowing other block or inline content inside depending on the element.

Examples: `<div>`, `<p>`, `<h1>`, `<section>`, and `<article>`.

Q5. Semantic HTML Elements

Semantic HTML elements clearly describe the meaning or purpose of their content. Examples include `<header>`, `<nav>`, `<main>`, `<section>`, `<article>`, `<aside>`, and `<footer>`.

They are important because they make HTML easier to understand and maintain, improve accessibility by giving assistive technologies meaningful structure, and can help search engines understand page organization.

Q6. Creating a Table Using HTML

The following HTML creates the requested table with headings, rows, and columns.

```
<table border="1">
  <tr>
    <th>Sr. No.</th>
    <th>Name</th>
    <th>Designation</th>
    <th>Department</th>
  </tr>
  <tr>
    <td>1</td>
    <td>Vijaya</td>
    <td>Manager</td>
    <td>Sales</td>
  </tr>
  <tr>
    <td>2</td>
    <td>Mayank</td>
    <td>Sr. Manager</td>
    <td>Marketing</td>
  </tr>
  <tr>
    <td>3</td>
    <td>Priyam</td>
    <td>Manager</td>
    <td>HR</td>
  </tr>
```

</table>

Q7. Creating a Nested List Using HTML

An unordered list can be nested inside an ordered or unordered list item. For the requested structure:

```
<ul>
  <li>River
    <ul>
      <li>Ganga</li>
      <li>Yamuna</li>
      <li>Brahmaputra</li>
      <li>Narmada</li>
    </ul>
  </li>
</ul>
```

Q8. Types of CSS

1. Inline CSS: CSS is written directly in an HTML element using the style attribute. Example: **<p style="color: blue;">Hello</p>**

2. Internal CSS: CSS is written inside a <style> element in the HTML document, usually inside <head>. Example: **<style> p { color: blue; } </style>**

3. External CSS: CSS is stored in a separate .css file and linked to the HTML page. Example: **<link rel="stylesheet" href="style.css">**. External CSS is useful for maintaining a consistent style across multiple pages.

Q9. Practical Understanding

HTML – Structure: HTML defines the content and structure of a webpage, such as headings, paragraphs, images, links, forms, and tables.

CSS – Presentation: CSS controls how that structure looks, including colors, fonts, spacing, layout, and responsiveness.

JavaScript – Behavior: JavaScript makes the webpage interactive and dynamic, for example by responding to clicks, validating forms, changing content, and fetching data.

Technology	Main role	Examples of contribution
HTML	Structure	Headings, text, images, forms, tables
CSS	Presentation	Colors, fonts, layout, responsive design
JavaScript	Behavior	Events, validation, dynamic updates, API calls

Q10. Short Answer

a. Role of Bootstrap: Bootstrap provides ready-made CSS classes and UI components, along with a responsive grid system, to help developers create consistent and mobile-friendly web interfaces quickly.

b. What is jQuery? jQuery is a JavaScript library designed to simplify common tasks such as selecting and modifying DOM elements, handling events, and creating animations. It became widely used because it reduced repetitive JavaScript code and helped address browser compatibility issues.

c. What is React.js? React.js is a JavaScript library for building user interfaces using reusable components. It helps solve the problem of managing complex, frequently changing interfaces by organizing the UI into components and efficiently updating the parts that need to change.